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Cross-Classified Random-Effects Difference-in-Differences Model: A Case Study for New York City's Congestion Pricing

--Manuscript Draft--

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ABSTRACT

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Novelty: A cross-classified random-effects difference-in-differences model is applied to each of the five modes, so substitution and suppression are compared at the same time and on comparable space, then decomposed by commute structure.

Practical Applications: Charges set by vehicle class should also consider a mode's exposure to the zone and its trip lengths. Where suppression is not absorbed by the untolloed modes, the outer ring and the inner rings off-peak, recycled toll revenue should add transit capacity in those two places rather than uniformly.

1 INTRODUCTION

On January 5, 2025, New York City activated the Central Business District Tolling Program (CBDTP), the first cordon-based congestion charge in the United States, over a Congestion Relief Zone (CRZ) covering the surface streets of Manhattan at or below 60th Street (Metropolitan Transportation Authority n.d.a). The charge is levied differently by vehicle class. Passenger and small commercial vehicles are tolled once per day on entering the CRZ, \$9 in the peak period and \$2.25 overnight with a valid E-ZPass, and repeated entries on the same day are charged only once. Vehicles licensed by the New York City Taxi and Limousine Commission are instead tolled per trip, on any trip that begins, ends or takes place entirely within the CRZ: \$0.75 for yellow taxis, green taxis and black cars, and \$1.50 for high-volume for-hire vehicles (FHV), chiefly Uber and Lyft (Metropolitan Transportation Authority n.d.a). Both per-trip charges are passed to the passenger.

This fee structure does not act on the transport system in a single direction. For the modes that bear the charge directly the cost of a trip rises. For the modes that do not, Subway and Citi Bike, the relative price falls. One policy therefore moves the relative prices of the two groups in opposite directions. Per-trip charging further makes the weight of the fee depend on trip length, since a fixed charge accounts for a larger share of the cost of a short trip than of a long one. The response also depends on other attributes of the trip. A morning inbound commute from the periphery into the CRZ and a short midday trip within the CRZ face different substitutes and different time constraints, and the same origin–destination (OD) path need not be equally substitutable in the inbound and the outbound direction.

This study estimates the effects of the policy's first year on the five modes with public trip-level OD records: Yellow Taxi, Uber, Lyft, Subway and Citi Bike. The unit of observation is a flow (a directed traffic stream in a given time window, fixed in Section 5.2 as one directed OD ring pair in one peak window). Estimating at the flow level puts the responses of the tolled and untolled modes on a common scale and lets them be read separately by travel direction, time of day and ring distance. The evaluations that precede this one are reviewed next, and the gaps they leave frame the design.

2 RELATED WORK

The theoretical basis of congestion pricing goes back to the pricing of traffic externalities (Vickrey 1969), and its first implementation was the area licensing scheme introduced in Singapore in 1975 (Watson and Holland 1976). Since the cordon charges adopted by London in 2003 and Stockholm in 2006, evaluations have reported substantial reductions in traffic within the charged zone (Leape 2006; Eliasson et al. 2009).

More recent evaluations extend the field of observation beyond the zone and to finer divisions of the traveling population. Herzog (2024) finds the effects of the London charge propagating city-wide through the road network rather than stopping at the cordon; Craik and Balakrishnan (2023) find by synthetic control that high-income travelers cut their chargeable trips without cutting total travel to the center, substituting toward untolled trips, while low-income travelers cut both and substituted far less; and the environmental and health effects of road pricing have been examined elsewhere (Green et al. 2020; Gibson and Carnovale 2015; Simeonova et al. 2021). The effects of a cordon charge are thus uneven in space and differ across population groups and outcome dimensions. These evaluations work mainly with private-vehicle volumes or zone-level totals.

The growth of ride-hailing altered both the composition of urban congestion and the travel structure on which a charge acts. Erhardt et al. (2019) attribute most of the growth in weekday vehicle hours of delay in San Francisco between 2010 and 2016 to transportation network companies (TNCs), with evidence in the same direction for US cities (Tarduno 2021), though Fageda (2021) finds the opposite for European cities, where Uber's entry is associated with lower congestion. Krishnamurthy and Ngo (2024) find the effect divided by time and density across California counties, so the net effect is not a single constant. Where taxis and ride-hailing hold an appreciable share of the vehicles in a charged zone, a charge on entry creates different price exposures across modes; competition between the New York platforms has been characterized separately (Rosaia 2025).

Whether the modes substitute for or complement one another is a persistent question. Hall et al. (2018) find Uber a complement to the average US transit agency, raising ridership about 5% two years after entry, more strongly in larger cities and for smaller agencies; Cats et al. (2022) show the two relations coexisting within one system, according to whether the trip has a time-competitive public transport alternative. For New York, ride-hailing, transit and their equity implications have been examined directly (Jin et al. 2019); on Manhattan and Brooklyn bus routes added Citi Bike docks accompany a decline in the route's daily unlinked trips (Campbell and Brakewood 2017); and the 2003 Los Angeles transit strike sharply raised average highway delay (Anderson 2014), indicating from another direction the volume of travel that transit absorbs. The balance thus depends on context.

New York offers one earlier episode. Li and Vignon (2024) evaluate the congestion surcharge levied from

February 2019 on taxi and ride-hailing trips in Manhattan below 96th Street and find those trips down by about 11% overall, concentrated on trips under one mile and in areas served by the Subway or Citi Bike, with no measurable improvement in traffic speeds. Two mechanisms follow: the bite of a charge depends on trip length, and on whether a substitute is available. Sector-specific charges in Chicago and San Francisco yield comparable findings (Zheng et al. 2023; Liang et al. 2023; Li et al. 2021).

The CBDTP prompted research before it took effect. Baghestani et al. (2020) predict by microsimulation that CRZ-related trips fall as the fee rises while trips internal to the zone do not, and Ghassabian et al. (2024) set out a framework for the health and health-equity consequences. Ostrovsky and Yang (2025) note that a fee structure differentiated by vehicle class places a markedly heavier burden on private drivers than on taxi and ride-hailing passengers; that price exposure differs by vehicle class also motivates estimating effects mode by mode below.

Since implementation, the policy has been assessed along several dimensions. On traffic operations and the environment, Cook et al. (2025) estimate by generalized synthetic control that speeds on CRZ roads rose by 11% and that driver welfare improved by at least \$14.3 million per week, almost all of it accruing to untolled trips outside the zone, while no detectable change was found in the air quality, retail transactions and foot traffic measured within the CRZ. For fine particulates at monitoring sites inside the CRZ, Fraser et al. (2025) report a 22% decline six months after implementation; the two studies differ in observational scope and pollutant measure. Wang and Di (2026) find no significant reduction in injuries or collisions, and Turkcan et al. (2026) construct a vehicle-density series from traffic cameras. On modal composition, Zhang et al. (2026) examine ride-hailing and the Subway with a two-way fixed-effects model and report Uber volume down by up to 6% and Subway ridership up by about 1%, with effects concentrated on short and low-fare trips, while Lyft rose over the same period under a rider credit offered by the platform.

On spatial distribution, Li et al. (2026) find bus and Subway ridership rising while total travel demand falls slightly, the transit gains extending beyond the Manhattan core, measured against a forecast counterfactual rather than an observed control group. Ren et al. (2026) estimate \$523 million per year in net toll revenue against \$397 million in accessibility losses, unevenly distributed across segments and origins, so that transit compensation is more cost-effective differentiated than uniform.

On method, difference-in-differences with heterogeneous treatment effects has been treated systematically (Callaway and Sant'Anna 2021; Baker et al. 2026); its central concern is the weighting bias of the two-way fixed-effects estimator under staggered adoption, which does not arise here because the CBDTP has a single treatment date. Synthetic control and its interactive fixed-effects extension relax the parallel-trends requirement (Abadie et al. 2010; Xu 2017); the CBDTP evaluations of Cook et al. (2025) and Shanbhag et al. (2026) follow that route, at the road-segment and neighborhood levels respectively. Hierarchical difference-in-differences structures, which let a treatment coefficient vary over a grouping structure, have been applied in other policy domains (Bosco et al. 2025) and are less common in transport policy evaluation.

Two gaps follow. Methodologically, a two-way fixed-effects design verifies parallel trends on one aggregate series while the effect is read far more finely, so the test is uninformative about the units at which the effect is read. Empirically, few studies place the tolled and the untolled modes on a common scale that separates time of day, space and direction at once. Against those gaps, the contribution of this work has three aspects.

Observational design. Five modes, three tolled and two untolled, are placed on one panel of directed OD ring pairs crossed with peak windows, on ring partitions that agree closely across modes, so that suppression and substitution across all five are measured on a single scale by one set of coefficients.

Estimation framework. A cross-classified random-effects structure, a hierarchical model whose two grouping factors cross rather than nest (Raudenbush 1993; Raudenbush and Bryk 2002), is embedded in difference-in-differences (DiD); one fit returns the pooled and the flow-level effects, and the pooled parallel-trends verdict resolves into one verdict per flow, so the interpretation rests only on the flows where the assumption holds.

Empirical findings. At the pooled level the suppression that is significant appears on Yellow Taxi and Uber. Resolved to flows, the commute structure organizes how the effects are distributed: untolled substitution is confined to the commute direction within Ring 1 and decays outward in the peak hours, whereas tolled suppression has no such spatiotemporal boundary and decays outward off-peak.

3 DATA

The five modes publish trip records in different formats and at different spatial granularities, so a common panel has to be built before any of them can be compared. **Figure 1** previews how the raw inputs are reorganized into the regression-ready panels, each block tagged with the subsection that details it.

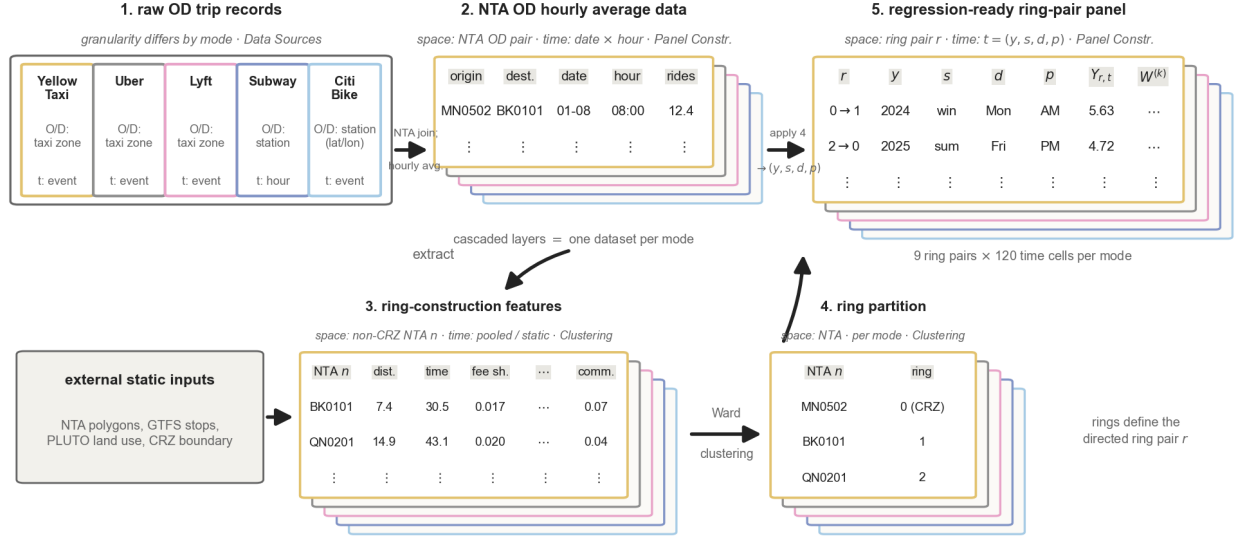


Figure 1 Data reorganization. Per-mode raw OD records (1), whose granularity differs by mode, are unified into Neighborhood Tabulation Area (NTA) level hourly averages (2) from the sources of Table 1. Ring-construction features (3, Table 2), extracted from (2) with the static inputs, yield the per-mode ring partition (4, Figure 2); applying it back to (2) gives the OD ring-pair panels (5, Table 3). Cascaded layers denote one dataset per mode, border colours identify the modes as in block 1, italic lines name the detailing subsection, and table entries are illustrative.

3.1 Data Sources

The core raw data of this study are the origin–destination (OD) ridership records of the five modes from January 2024 to December 2025, supplemented by New York City weather data, the land-use composition shares of each NTA, and several boundary and geographic reference layers. **Table 1** summarizes the source, type, role and coverage of every dataset.

3.2 Neighborhood-Based Clustering

A binary split into charged and uncharged neighborhoods would not show how the effect changes with distance from the zone, and existing NYC evidence reports exactly such spatial variation (Zhang et al. 2026; Li et al. 2026). The relevant distance is how far a neighborhood sits from the CRZ in travel terms rather than geometric ones, and it differs by mode: one well connected to the CRZ by rail need not be well connected by bicycle. The non-CRZ neighborhoods are therefore grouped into a near and a far ring, separately for each mode and from that mode’s own travel characteristics.

For each non-CRZ NTA, ten features are computed within mode from 2023–2024 trips, the CRZ fee share from the 2025 schedule, and z-scored (**Table 2**): four capture its travel linkage with and travel cost to the 17 CRZ NTAs, five describe transit accessibility and land use, and the CRZ fee share records the direct policy exposure of its trips, dropped for the untolled Subway and Citi Bike, which leaves nine features for those two. Each served non-CRZ NTA is represented by that feature vector and partitioned by Ward’s agglomerative hierarchical clustering (Virtanen et al. 2020) on pairwise Euclidean distances, cut into two clusters.

The result is that every NTA carries one of three labels per mode. Ring 0 is the CRZ interior, the 17 Manhattan NTAs fixed by the policy boundary. Of the two clusters, the one with the lower mean travel distance to the CRZ is labeled Ring 1 (near) and the other Ring 2 (far). NTAs outside a mode’s service area remain unassigned.

The ring maps (**Figure 2**) show a coherent structure. The tolled modes yield nearly identical partitions, Uber and Lyft almost indistinguishably so. Subway and Citi Bike instead follow their own station geography over a smaller service area. All five modes share a 32-NTA Ring 1 core around the CRZ and a 27-NTA Ring 2 core beyond it, and cross-mode disagreement is confined to the 20-NTA transition band between them, so the spatial assignment is largely common to the five modes.

TABLE 1 Data Sources; Each Row Cites Its Own Dataset. TLC, MTA and DCP Are the New York City Taxi and Limousine Commission, the Metropolitan Transportation Authority and the New York City Department of City Planning. Panels Cover January 2024 to December 2025, and Static Layers Carry No Time Dimension

dataset	source	type	role
docked-bike OD trip records	Citi Bike (n.d.)	OD (trip) panel	Citi Bike ring-construction features and panel response $Y_{r,t}$
Yellow-taxi and high-volume FHV (Uber, Lyft) trip records	TLC (n.d.a)	OD (trip) panel	the same two roles for Yellow Taxi, Uber and Lyft; fare fields give the CRZ fee share feature
taxi-zone shapefile	TLC (n.d.b)	static	joins every trip endpoint to its 2020 NTA
station-pair subway OD ridership	MTA (2023; 2024; 2025)	OD (station) panel	the same two roles for the Subway
subway and bus GTFS feeds	MTA (n.d.b)	static	subway distance and time features; station and bus-stop density
CBDTP definition of the CRZ	MTA (n.d.a)	static	fixes Ring 0 = the 17 Manhattan NTAs at or below 60th Street
2020 NTA polygons	DCP (2020)	static	the spatial unit throughout: partition, maps, join target, land areas, weather polygons
PLUTO tax-lot land-use shares	DCP (n.d.)	static	residential and commercial area shares among the ring features
hourly weather on 17 grid cells	Open-Meteo (2023)	grid hourly panel	the five candidate weather covariates
U.S. federal and NY State holiday calendar	Vacanza (2025)	static	defines the workday calendar; holidays are dropped before every workday aggregate

TABLE 2 Ring-Construction Features of a Non-CRZ NTA n , z-Scored Within Mode over Its CRZ-Connected OD Pairs $\mathcal{E}_n$; q_e , dist_e , time_e , cost_e , fee_e Are the Rides, Distance, Time, Fare and CRZ Charge of Pair e , and area_n the DCP Polygon Area

group	name	definition	dimension	period
Cost	travel distance	$\sum_{e \in \mathcal{E}_n} q_e \text{dist}_e / \sum_{e \in \mathcal{E}_n} q_e$ (volume-weighted over CRZ-connected trips)	miles	2023–2024
Cost	travel time	$\sum_{e \in \mathcal{E}_n} q_e \text{time}_e / \sum_{e \in \mathcal{E}_n} q_e$	minutes	2023–2024
Policy	CRZ fee share	$\sum_{e \in \mathcal{E}_n} \text{fee}_e / \sum_{e \in \mathcal{E}_n} \text{cost}_e$ (tolled modes only; dropped for Subway and Citi Bike)	fraction	2025
Linkage	CRZ-connected rides	$\log(1 + \sum_{e \in \mathcal{E}_n} q_e)$	dimensionless	2023–2024
Linkage	CRZ reach breadth	$\#\{\text{distinct CRZ NTAs exchanged with}\}$	count (0–17)	2023–2024
Accessibility	subway, bus and Citi Bike density (three features)	$\#\{\text{stations or stops of each type in } n\} / \text{area}_n$	stations/km ²	static
Land use	residential and commercial share (two features)	$\text{area}_n^{\text{res}} / \text{area}_n$ and $\text{area}_n^{\text{comm}} / \text{area}_n$	fraction	static

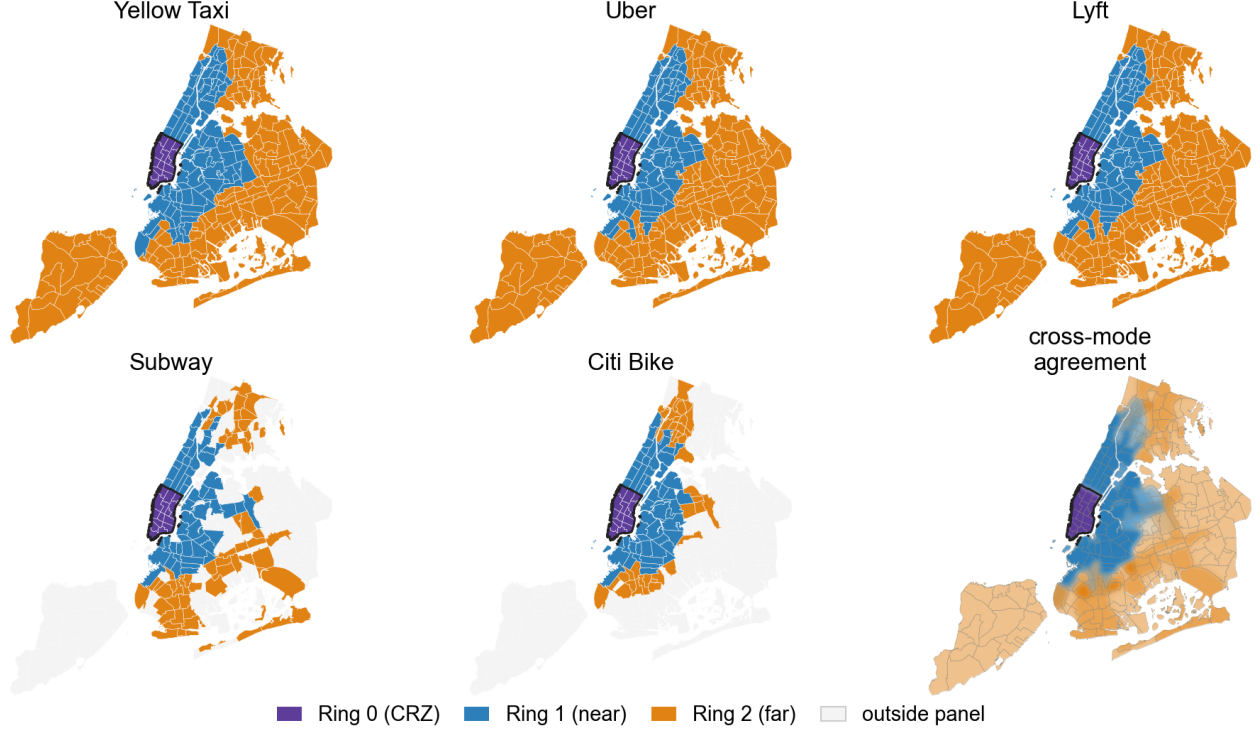


Figure 2 Per-mode ring partition; black outline = CRZ. Last panel: each NTA tinted by the share of the five modes assigning it to each ring.

3.3 Model-Ready Panel Data Construction

The per-mode ring partitions are applied back to the NTA-level OD hourly panels to build the estimation sample. Peak windows are fixed at 07:00–09:00 (morning peak) and 17:00–19:00 (evening peak), with the remaining hours pooled as off-peak, and each mode’s trips are aggregated to the OD ring pair $\times t = (y, s, d, p)$ time index: nine directed OD ring pairs observed over year $\times$ season $\times$ day-of-week $\times$ peak window. Averaging over workdays before taking logs keeps entries comparable across seasons and days-of-week, whose workday counts differ. **Table 3** lists the resulting panel variables, the additive main effects and multiplicative interaction (AMMI) covariate among them, constructed in Section 5.3.

Response. For each OD ring pair r and time index $t = (y, s, d, p)$, ridership is measured as the workday daily average,

$$Y_{r,t}^{(m)} = \log(1 + \overline{\text{rides}}_{r,t}^{(m)}), \quad (1)$$

$$\overline{\text{rides}}_{r,t}^{(m)} = \frac{\text{total rides on all OD trips within } r \text{ during peak } p}{\#\{\text{workdays of } (y, s, d)\}}. \quad (2)$$

The $\log(1 + \cdot)$ transform lets every DiD coefficient be read as a percentage change, $\exp(\beta) - 1$, which is how every treatment effect is reported from here on.

Treatment definition. Treatment is defined by geographic exposure to the CRZ: an OD ring pair is treated if it touches Ring 0, the CRZ interior,

$$\text{Treated}_r^{(m)} = \mathbf{1}(0 \in r), \quad (3)$$

where Ring 0 is fixed by the CBDTP boundary. A directed pair from Ring i to Ring j is written $i \rightarrow j$ from here on. Five of the nine pairs are treated; the four among Rings 1 and 2 serve as controls. Given implementation on January 5, 2025, the post-toll period indicator is

$$\text{Post}_y = \mathbf{1}(y \geq 2025), \quad (4)$$

and the time-varying treatment indicator entering the DiD term is the product $\text{Post}_y \text{Treated}_r^{(m)}$.

TABLE 3 Panel Variables; the Mode Superscript (m) Is Dropped, and the Covariate Rows List the Full Candidate Pool $\mathcal{S}$. Every Weather Candidate Is Resolved to the Ring Pair and Is Constant over Day-of-Week

symbol	definition	levels / dimension
m	mode (fit separately per mode): the three tolled modes — Yellow Taxi (\$0.75 CRZ charge per trip) and Uber, Lyft (\$1.50 per trip) — and the two untolled modes Subway and Citi Bike	5 (3 tolled + 2 untolled)
r	ring pair, directed Ring $i \rightarrow$ Ring j (0 = CRZ, 1 = near, 2 = far); treated iff it touches Ring 0	9 (5 treated, 4 control)
$t = (y, s, d, p)$	time index = year $\times$ season $\times$ day-of-week $\times$ peak	120
y	year	{2024, 2025}
s	season	4 (winter, spring, summer, fall)
d	day-of-week (workdays)	5 (Mon–Fri)
p	peak: morning peak = hours 07–09, evening peak = hours 17–19, off-peak = all other hours	3
$Y_{r,t}$	response, $Y_{r,t} = \log(1 + \text{avg daily rides})$, where avg daily rides = (total rides on all OD trips within r during peak p) / (# workdays of (y, s, d))	log rides
$W_{r,t}^{(1)}$	temperature 2 m; hourly values, averaged over the hours of peak p , then z -scored	dimensionless (raw °C)
$W_{r,t}^{(2)}$	wind gusts 10 m; hourly values, averaged over the hours of peak p , then z -scored	dimensionless (raw km/h)
$W_{r,t}^{(3)}$	relative humidity 2 m; hourly values, averaged over the hours of peak p , then z -scored	dimensionless (raw %)
$W_f^{(4)}$	AMMI covariate; the peak-direction score of a flow, built on the dimension f fixed in Section 5.3	dimensionless
$W_{r,t}^{(5)}$	rain (past hour); hourly values, averaged over the hours of peak p , then z -scored; candidate, not selected	dimensionless (raw mm)
$W_{r,t}^{(6)}$	snowfall (past hour); hourly values, averaged over the hours of peak p , then z -scored; candidate, not selected	dimensionless (raw cm)

4 METHODOLOGY

Three difference-in-differences models are compared on those panels, differing only in how the treatment coefficient is parameterized; this section defines them and the procedure that fits each.

4.1 Model Family

Three candidate models are considered, all fit on the per-mode panels of **Table 3** and forming a progression in how the treatment effect is parameterized. M0 estimates a single pooled coefficient and reports how much ridership changed on average, without resolving where or when. M1 resolves that coefficient onto a spatiotemporal dimension and recovers the heterogeneity, at the cost of thinner samples per coefficient and of no longer reporting an overall effect, the two being collinear and so not carryable together in one fixed-effect model. M2 instead places both the crossed intercepts and the DiD slope in a random-effects structure, so that a pooled effect and its flow-level departures are estimated jointly.

The three share the same response and covariate structure, so the comparison reported later reflects the parameterization of the DiD term alone. Each model is fit separately per mode; every quantity is mode-specific, and the superscript (m) is dropped where only one mode is in view. The response is conditionally Gaussian,

$$Y_{r,t}^{(m)} \sim \mathcal{N}(\mu_{r,t}^{(m)}, \sigma^{2(m)}), \quad (5)$$

with residual $\varepsilon_{r,t}^{(m)} = Y_{r,t}^{(m)} - \mu_{r,t}^{(m)}$. The precision controls are the candidate set $\mathcal{S} = \{1, \dots, J\}$ of covariates $W^{(m,1)}$ to $W^{(m,J)}$, each carrying one coefficient $\beta_{k+1}^{(m)}$.

M0, the two-way fixed-effects (TWFE) DiD:

$$Y_{r,t}^{(m)} = \lambda_r^{(m)} + \lambda_t^{(m)} + \beta_1^{(m)} \text{Post}_y \text{Treated}_r^{(m)} + \sum_{k \in S} \beta_{k+1}^{(m)} W_{r,t}^{(m,k)} + \varepsilon_{r,t}^{(m)}. \quad (6)$$

M1, the interacted TWFE DiD, in which the DiD coefficient varies along the flow dimension c :

$$Y_{r,t}^{(m)} = \lambda_r^{(m)} + \lambda_t^{(m)} + \beta_{1,c}^{(m)} \text{Post}_y \text{Treated}_r^{(m)} + \sum_{k \in S} \beta_{k+1}^{(m)} W_{r,t}^{(m,k)} + \varepsilon_{r,t}^{(m)}. \quad (7)$$

M2, the cross-classified random-effects DiD. Here the two grouping factors are the OD ring pair r and the time index t : every ring pair is observed at every time index and the reverse, so the factors cross rather than nest, which is what the name records (Raudenbush 1993; Raudenbush and Bryk 2002). Each carries its own random intercept, u_r and u_t , and these play the same identifying role as the fixed effects λ_r and λ_t of M0 and M1, absorbing heterogeneity unrelated to the policy. The DiD slope in addition carries a random effect on the crossed dimension c , so one model delivers both the pooled coefficient $\beta_1^{(m)}$ and the flow-level coefficients $\beta_1^{(m)} + u_c^{(m)}$. Level 1, the conditional mean:

$$\mu_{r,t}^{(m)} = \beta_{0,r,t}^{(m)} + \beta_{1,c}^{(m)} \text{Post}_y \text{Treated}_r^{(m)} + \sum_{k \in S} \beta_{k+1}^{(m)} W_{r,t}^{(m,k)}. \quad (8)$$

Level 2, the varying intercept and the varying slope:

$$\beta_{0,r,t}^{(m)} = \beta_0^{(m)} + u_r^{(m)} + u_t^{(m)}, \quad \beta_{1,c}^{(m)} = \beta_1^{(m)} + u_c^{(m)}, \quad (9)$$

$$u_r^{(m)} \sim \mathcal{N}(0, \sigma_r^{2(m)}), \quad u_t^{(m)} \sim \mathcal{N}(0, \sigma_t^{2(m)}), \quad u_c^{(m)} \sim \mathcal{N}(0, \sigma_c^{2(m)}). \quad (10)$$

Combined:

$$\mu_{r,t}^{(m)} = \beta_0^{(m)} + u_r^{(m)} + u_t^{(m)} + (\beta_1^{(m)} + u_c^{(m)}) \text{Post}_y \text{Treated}_r^{(m)} + \sum_{k \in S} \beta_{k+1}^{(m)} W_{r,t}^{(m,k)}. \quad (11)$$

Table 4 collects the symbols of the three models, their coefficients, dimensions, covariate sets, random effects and residual, with the role each plays.

4.2 Estimation

M0 and M1 are linear fixed-effect models whose grouping factors, the OD ring pair and the year–season–day-of-week index, enter the design matrix as indicator variables, so both are fit by ordinary least squares with `lm` in R (R Core Team 2025), with no within transformation or absorbing step. Each flow-level coefficient of M1 is a separate least-squares estimate sharing no information with the others.

For M2 there is no closed-form least-squares solution, its DiD slope being a random effect whose variance is itself unknown: the variance components are found by numerical optimization, with the fixed effects profiled out at every trial value. Algorithm 1 sets out the procedure.

Algorithm 1 Restricted maximum likelihood (REML) fitting of M2, for one mode

Require: panel $\{Y_{r,t}, W_{r,t}, \text{Post}_y, \text{Treated}_r\}$; crossed factors r and t ; DiD dimension c

Ensure: pooled and flow-level coefficients; variance components

- 1: **Assemble the design.** Fixed part $X = [\mathbf{1}, \text{Post}_y \text{Treated}_r, W]$; random part Z carrying the crossed intercepts u_r , u_t and the DiD slope on c
 - 2: **Parameterize the variance structure.** Write $\text{Var}(b) = \sigma^2 \Lambda_\theta \Lambda_\theta^\top$, in which θ holds the variances relative to σ^2
 - 3: **Solve the penalized least-squares problem at a trial θ .** $\min_{\beta, b} \|y - X\beta - Z\Lambda_\theta b\|^2 + \|b\|^2$, by sparse Cholesky, returning $\hat{\beta}(\theta)$ and $\hat{b}(\theta)$ and so profiling β and σ^2 out of the likelihood
 - 4: **Evaluate the criterion.** Compute the profiled REML deviance $d_R(\theta)$, whose term $|X^\top V^{-1} X|$ corrects the downward bias of the maximum-likelihood variance estimates
 - 5: **Optimize the variance components.** Update θ by bobyqa, a derivative-free trust-region search, and repeat the previous two steps until d_R converges
 - 6: **Read off both levels.** At $\hat{\theta}$ take the conditional modes $\hat{b}$: the pooled effect is $\hat{\beta}_1$ and the flow-level effects are $\hat{\beta}_1 + \hat{u}_c$, the best linear unbiased predictors (BLUPs)
-

The algorithm is implemented by the R package `lme4` (Bates et al. 2015) under the REML criterion, with the `bobyqa` optimizer and a limit of 2×10^5 function evaluations; the Satterthwaite degrees of freedom carried by the fixed

TABLE 4 Symbols of M0, M1, and M2: Coefficients, Dimensions, Covariate Sets, Random Effects and Residual

symbol	name	interpretation / meaning
β_0	grand intercept	$\exp(\beta_0) \approx$ baseline daily rides (M2)
$\beta_{0,r,t} = \beta_0 + u_r + u_t$	varying intercept	M2 only
β_1	pooled DiD coefficient	$\exp(\beta_1) - 1 =$ pooled % change in daily ridership attributable to the toll
c, f	DiD and AMMI dimensions	products of panel factors along which the DiD coefficient and the AMMI covariate vary; one cell of c is a flow. Both fixed in Sections 5.2 and 5.3
$\beta_{1,c}$	flow-level DiD coefficient	$\exp(\cdot) - 1 =$ % change for flow c . M1: a fixed effect per flow; M2: $\beta_{1,c} = \beta_1 + u_c$ (pooled effect + random slope)
$S = \{1, \dots, J\}, \mathcal{D} \subset S$	candidate sets	S indexes the full covariate pool of Table 3, $\mathcal{D}$ its five weather members
$\mathcal{D}^*, S^*$	adopted sets	the weather terms kept, and those together with the AMMI covariate; S^* is carried by the final fit
$\beta_{k+1}, k \in S$	covariate coefficients	$\exp(\beta_{k+1}) - 1 =$ % change per +1 standard deviation (SD) of a weather term; the AMMI coefficient is read on the log scale, per +1 unit of the peak-direction score
λ_r, λ_t	ring-pair / time fixed intercepts	M0/M1
u_r	ring-pair intercept random effect	mean zero (M2)
u_t	time intercept random effect	mean zero (M2)
u_c	flow slope random effect	mean zero (M2)
$\sigma_r^2, \sigma_t^2, \sigma_c^2$	variance components	one per random effect
$\mu_{r,t}$	conditional mean	$Y_{r,t} \sim \mathcal{N}(\mu_{r,t}, \sigma^2)$ (log scale)
$\varepsilon_{r,t} = Y_{r,t} - \mu_{r,t}$	residual	carries no crossed intercept, DiD, weather or AMMI term
σ^2	residual variance	Gaussian observation layer, $(\log)^2$

effects come from lmerTest (Kuznetsova et al. 2017). The random effects and the residual are taken to be independent. The difference is one of procedure, not only of specification. M0 returns the pooled coefficient alone and M1 the flow-level ones alone, each estimated on its own; M2 returns both, and its flow-level effects are conditional modes of a random slope, drawn toward the pooled coefficient by an amount set by the estimated variance ratio, so a flow observed on few rides borrows strength from the rest instead of standing on its own thin sample. And because both levels come out of one criterion in one fit, parallel trends can be tested at each against the same estimated object.

5 VARIABLE SELECTION

Section 4 fixed the skeleton the three models share: the terms that absorb heterogeneity unrelated to the policy, fixed effects in M0 and M1 and the crossed random intercepts that correspond to them in M2, and the DiD interaction that identifies the policy. Three components of that skeleton were left open, each a selection with its own criterion. The first is the precision controls: which members of the candidate set S to retain, a member qualifying when it reduces residual variance without moving the treatment estimate. The second is the dimension c the DiD coefficient varies along, chosen where the heterogeneity of the effect is strongest and so most worth resolving. The third follows from the second: once the coefficient is resolved onto c , the interaction already present in the pre-period ridership surface between place and time of day is absorbed by those same coefficients, so a covariate is needed to hold it out, along with its own dimension f .

Choosing the controls, c and f is not separable, since each is defined on models carrying the other two, so the three are run coordinate-wise, each experiment fixing the other two at their adopted values. The reported triple $(\mathcal{D}^*, c, f)$ is a fixed point: re-running any one selection conditional on the other two reproduces the same choice, and each choice also survives model variants that alter the remaining components, so the order of selection does not matter. Each subsection below states its adopted choice and the premise on which it is conditioned.

5.1 Weather Covariate Selection

The weather members of $\mathcal{S}$ form the subset $\mathcal{D}$, the five variables listed in **Table 3**. On M0, M1 and M2, with the DiD dimension fixed at its adopted value $c = r \times p$ (Section 5.2) and the AMMI covariate entering on the matched dimension (Section 5.3), all $2^5 = 32$ subsets of $\mathcal{D}$ are refit by maximum likelihood (REML likelihoods are not comparable across fixed-effect structures) and compared by the Akaike information criterion (AIC) averaged over the five modes. A candidate is kept if dropping it worsens the mean AIC on every model, that is, if its mean leave-one-out ΔAIC is positive throughout (**Table 5**). Wind gusts, relative humidity and temperature qualify; rain and snowfall are AIC-neutral and dropped, giving $\mathcal{D}^* = \{\text{temperature, wind gusts, relative humidity}\}$.

Identification is insensitive to this choice. Each of the 32 subsets of $\mathcal{D}$ was compared with the no-weather model by Welch equality tests on the DiD coefficients themselves, and no difference is significant, $p \geq 0.81$ pooled and $p \geq 0.28$ across all 2,325 flow-level comparisons, so the weather terms absorb residual variance without moving what is later interpreted.

TABLE 5 Weather Covariate Selection: Mean Leave-One-Out ΔAIC over the Five Modes When Each Candidate Is Dropped, on All Three Models (ML Refits; Positive = Dropping Worsens Fit)

candidate covariate	mean ΔAIC when dropped (ML refits)			net	decision
	M0	M1	M2		
temperature	+1.3	+2.2	+1.9	+5.4	keep
rain	-0.6	-0.3	-0.3	-1.2	drop
snowfall	+0.2	+0.1	-0.3	-0.1	drop
wind gusts	+9.6	+5.3	+4.0	+18.8	keep
relative humidity	+2.3	+2.4	+3.1	+7.8	keep

5.2 Selection of the DiD Interaction Dimension

On M1 and M2 the DiD interaction dimension is set to the fully interacted $c = r \times s \times p \times d$, where the season is coarsened to three levels (winter, summer and the rest, spring plus fall), resolving the DiD coefficient into $5 \times 3 \times 3 \times 5 = 225$ groups. The year is not a candidate dimension: Post_y is itself defined on the year, so a DiD coefficient varying by year would not be separately identified. The weather covariates are fixed at the set adopted in the weather selection, and the AMMI covariate enters on the dimension matched to it, the matching requirement being established in Section 5.3. The 225 coefficients are then decomposed by a balanced four-way analysis of variance (ANOVA), which splits their total variation into fifteen orthogonal sums of squares (four main effects, six two-way, four three-way and the four-way term); each component's share of heterogeneity is its sum of squares as a percentage of the total. The aggregated blocks reassign each interaction term equally to its parent dimensions, a two-way term half to each and so on, so they also sum to 100% (**Table 6**).

The OD ring pair $\times$ peak window interaction dominates on both models while season and day-of-week are marginal, so the adopted dimension is $c = r \times p$, resolving the DiD coefficient into $5 \times 3 = 15$ coefficients, one per treated flow. This fixes the flow for the rest of the paper: one directed OD ring pair observed in one peak window.

5.3 Selection of the AMMI Covariate

The pre-period ridership surface carries an interaction between where a trip runs and when it runs that has nothing to do with the toll. Left in the residual, it is absorbed by the flow-level DiD coefficients, which are defined on the same two dimensions. The AMMI covariate holds that structure out of the treatment estimate at the cost of one coefficient.

The AMMI covariate is accordingly built by the decomposition of Gauch (1988). On a row dimension a and a column dimension b , which together form $f = a \times b$, averaging the pre-period (2024) response over seasons and

TABLE 6 DiD Interaction-Dimension Selection: ANOVA Decomposition of the 225 Fully Interacted DiD Coefficients, Mean over Modes, on M1 (Fixed Effects) and on M2; Components Abbreviated by Their Dimension Products (e.g., $rp = r \times p$; seas. = Season, dow = Day-of-Week)

share (%)	orthogonal components															aggregated blocks			
	r	s	p	d	rs	rp	rd	sp	sd	pd	rsp	rsd	$rp d$	spd	$rsp d$	r pair	peak	seas.	dow
M1	8.1	8.2	6.5	7.0	6.1	38.4	8.6	2.1	0.8	3.6	0.6	0.7	8.1	0.5	0.6	37.9	31.8	13.5	16.8
M2	5.5	4.7	2.6	4.0	7.5	47.0	10.7	1.0	0.6	2.1	1.6	1.1	10.1	0.4	1.1	42.7	31.9	10.5	14.8

days-of-week gives the mean matrix $M_{a,b}^{(m)}$, which covers all nine ring pairs (9×3 for $r \times p$). Double-centring leaves only its interaction,

$$\tilde{M}_{a,b} = M_{a,b}^{(m)} - \bar{M}_a - \bar{M}_b + \bar{M}. \quad (12)$$

Its singular value decomposition,

$$\tilde{M} = \sum_k \omega_k g_k h_k^T, \quad (13)$$

with singular values $\omega_1 \geq \omega_2 \geq \dots$ and unit-norm vectors g_k, h_k , yields the leading pair: the row score $g^{(m)} = g_1$ and the column score $h^{(m)} = h_1$, both signs flipped if needed so that the largest column score is positive. Their product is the AMMI covariate,

$$W_{a,b}^{(m,4)} = g_a^{(m)} h_b^{(m)}, \quad (14)$$

one value per (a, b) combination, merged onto every panel row carrying it (numbered as in **Table 3**).

Whether f must match c is then settled by experiment. The DiD dimension is held at its adopted value $c = r \times p$ and only f is varied, so five candidates are compared: one built on the same OD ring pair $\times$ peak window dimension as the DiD coefficient and therefore matching it, three built on other dimensions (OD ring pair $\times$ season, OD ring pair $\times$ day-of-week, peak window $\times$ season), and a fifth carrying no AMMI covariate at all. Each is fit both on M1 (Equation 7, fixed-effect coefficients) and on M2, with a random DiD slope on the same dimension.

Figure 3 compares the five specifications on both models. Only the dimension-matched AMMI covariate holds the 75 flow-level estimates within a plausible range: its cross-flow standard deviation is 13.1% on M1 and 12.1% on M2, against roughly 37–40% for the other four, under which several flows imply changes beyond $\pm 60\%$. The coefficient β_5 is the change in log ridership per unit of the peak-direction score; because that score is built from pre-period data alone, the commute structure it carries is unrelated to the policy.

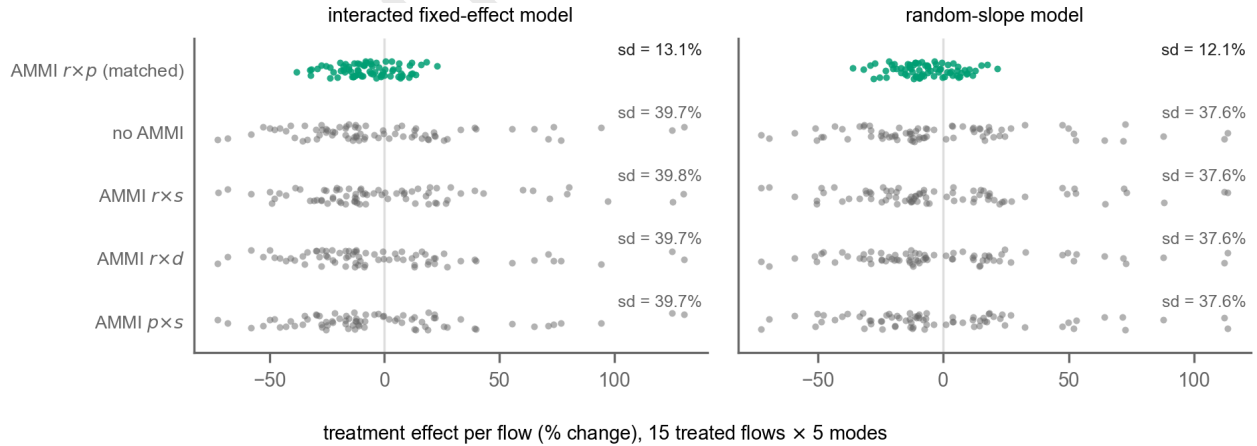


Figure 3 AMMI covariate selection. Flow-level DiD estimates of the 15 treated flows in each mode under the five candidate AMMI covariates, on the interacted fixed-effect model (left) and the random-slope model (right); the number beside each row is that row's cross-flow standard deviation.

6 MODEL SELECTION

With the three components settled in Section 5 now fixed, the covariate set $\mathcal{S}^*$, the DiD dimension $c = r \times p$ and the AMMI dimension $f = c$, the three models differ only in how the DiD term is parameterized. M2 nests by construction what M0 and M1 report separately, and whether that nesting holds empirically is tested by two-sided Welch–Satterthwaite equality tests against the model that targets each quantity, M0’s β_1 for the pooled effect and the 15 fully interacted flow-level coefficients for the flow-level effects. **Table 7** reports the outcome: across the five modes, none of the 16 comparisons per mode is rejected at 5%. M2 therefore reproduces both sets within sampling error and is adopted.

TABLE 7 Model Selection: p -Values of Two-Sided Welch–Satterthwaite Equality Tests. Pooled Column, β_1 from M0 vs from M2; Flow Columns, Each Fully Interacted Flow-Level DiD vs the M2 Predictor $\beta_1 + u_{r,p}$; Last Column Counts the 16 Tests Not Rejected at 5%

mode	pooled	0→0 morning peak	0→0 off-peak	0→0 evening peak	0→1 morning peak	0→1 off-peak	0→1 evening peak	0→2 morning peak	0→2 off-peak	0→2 evening peak	1→0 morning peak	1→0 off-peak	1→0 evening peak	2→0 morning peak	2→0 off-peak	2→0 evening peak	not sig. diff.
Yellow Taxi	0.585	0.553	0.374	0.930	0.686	0.643	0.971	0.576	0.694	0.888	0.865	0.531	0.891	0.674	0.599	0.907	16/16
Uber	0.791	0.972	0.875	0.912	0.950	0.942	0.974	0.858	0.833	1.000	0.999	0.903	0.974	0.893	0.931	0.987	16/16
Lyft	0.940	0.712	0.947	0.821	0.815	0.959	0.906	0.893	0.964	0.872	0.797	0.934	0.835	0.862	0.973	0.959	16/16
Subway	0.684	0.547	0.775	0.637	0.707	0.804	0.715	0.875	0.891	0.990	0.667	0.849	0.704	0.878	0.878	0.985	16/16
Citi Bike	0.422	0.579	0.399	0.983	0.719	0.501	0.918	0.664	0.606	0.886	0.756	0.547	0.968	0.609	0.616	0.830	16/16

The model taken to the final fit is therefore M2 with $\mathcal{S}^* = \{1, 2, 3, 4\}$ (temperature, wind gusts, relative humidity and the AMMI covariate, numbered as in **Table 3**) and $c = f = r \times p$. Substituting these into the two levels of M2 in Section 4 gives the adopted specification:

$$\mu_{r,t}^{(m)} = \beta_0^{(m)} + u_r^{(m)} + u_t^{(m)} + (\beta_1^{(m)} + u_{r,p}^{(m)}) \text{Post}_y \text{Treated}_r^{(m)} + \sum_{k \in \mathcal{S}^*} \beta_{k+1}^{(m)} W_{r,t}^{(m,k)}. \quad (15)$$

The fourth covariate $W^{(m,4)}$ is the AMMI score, written with subscript r, t like the others though it varies on $f = r \times p$.

7 EXPERIMENTS AND RESULTS

The adopted specification is fit once per mode, its identifying assumption is checked, and the estimates are then read at the pooled level and flow by flow.

7.1 Final Selected-Model Fitting

Table 8 reports the fixed effects with 95% confidence intervals and Satterthwaite p -values, together with the variance components and fit statistics. The 15 flow-level coefficients $\beta_1 + u_{r,p}$ of each mode are carried into Figures 4 and 5.

7.2 Parallel-Trends Checking

Parallel trends is the assumption on which identification in a difference-in-differences design rests: absent the toll, ridership on treated and control flows would have evolved along a common path, so that any post-period divergence between them is attributable to the policy. Zhang et al. (2026) assess it in a comparable NYC setting by two-way demeaning the outcome, forming the treated-minus-control mean residual difference, and testing whether its pre-period mean is zero. The residual-difference slope test (RDST) used here retains that construction and shifts the target from the level of the difference to its drift. It is run on the M2 fit, once per treated flow and once pooled. The M2 residuals $\varepsilon_{r,t}^{(m)} = Y_{r,t}^{(m)} - \hat{\mu}_{r,t}^{(m)}$ carry no crossed intercepts, no DiD term, no weather and no AMMI. The control mean is subtracted from the treated residual,

$$dR_{r;y,s,d}^{(m)} = \varepsilon_{r;y,s,d}^{(m),\text{treated}} - \bar{\varepsilon}_{y,s,d}^{(m),\text{ctrl}}, \quad (16)$$

TABLE 8 Full M2 Results: Fixed Effects (with 95% Confidence Intervals (CI), Satterthwaite df, and p) and Variance Components with Model Fit; the Flow-Level Coefficients $\beta_1 + u_{r,p}$ Are Shown in Figure 4

	Yellow Taxi	Uber	Lyft	Subway	Citi Bike
<i>fixed effects</i>					
β_0 grand intercept	7.459	9.150	8.101	11.495	6.685
95% CI	[6.476, 8.443]	[8.440, 9.860]	[7.390, 8.812]	[10.978, 12.011]	[4.862, 8.508]
p	<0.001	<0.001	<0.001	<0.001	<0.001
β_1 pooled DiD	-0.226	-0.064	-0.052	-0.012	-0.037
95% CI	[-0.300, -0.153]	[-0.128, -0.000]	[-0.132, 0.028]	[-0.075, 0.052]	[-0.101, 0.026]
p	<0.001	0.049	0.186	0.700	0.236
β_2 temperature	-0.015	-0.025	-0.034	-0.022	0.334
95% CI	[-0.161, 0.131]	[-0.150, 0.100]	[-0.163, 0.095]	[-0.117, 0.073]	[0.232, 0.436]
p	0.840	0.695	0.604	0.643	<0.001
β_3 wind gusts	-0.265	-0.029	-0.146	-0.051	-0.093
95% CI	[-0.416, -0.114]	[-0.142, 0.083]	[-0.270, -0.021]	[-0.127, 0.025]	[-0.211, 0.026]
p	<0.001	0.609	0.022	0.187	0.125
β_4 relative humidity	-0.130	-0.088	-0.151	-0.063	-0.136
95% CI	[-0.260, -0.001]	[-0.184, 0.007]	[-0.257, -0.044]	[-0.120, -0.007]	[-0.245, -0.026]
p	0.048	0.068	0.006	0.029	0.015
β_5 AMMI	0.926	0.741	0.741	2.263	1.988
95% CI	[0.873, 0.979]	[0.700, 0.783]	[0.694, 0.789]	[2.226, 2.300]	[1.938, 2.038]
p	<0.001	<0.001	<0.001	<0.001	<0.001
<i>variance components and fit</i>					
σ_r^2 ring-pair intercept	1.621	0.842	0.844	0.445	5.619
σ_t^2 time intercept	0.616	0.455	0.483	0.266	0.297
σ_{rp}^2 flow slope	0.014	0.012	0.019	0.012	0.011
σ^2 residual	0.020	0.009	0.012	0.008	0.014
AIC	-342.9	-1094.7	-872.0	-1282.5	-741.8
BIC	-293.1	-1044.9	-822.1	-1232.7	-692.0

1 where $\bar{\varepsilon}_{y,s,d}^{(m),\text{ctrl}}$ is the mean over the four control OD ring pairs. The resulting series is regressed on the calendar season
2 index $\tau = 1, \dots, 8$ (2024 seasons 1–4, 2025 seasons 5–8),

$$3 \quad dR = \alpha + \gamma \tau + \delta \text{Post} + v_{ys} + \eta, \quad (17)$$

4 in which α is the intercept, γ the drift per season, δ a post-period level shift and η the residual. The year-season
5 random intercept $v_{ys} \sim \mathcal{N}(0, \sigma_v^2)$ absorbs what the workdays of a given year and season share, so that the several
6 days-of-week observed inside one season are not independent evidence of a trend. A flow passes at the 5% level when
7 a Satterthwaite t -test does not reject $H_0: \gamma = 0$. The pooled test applies the same regression to one series, the treated
8 flows' dR averaged with weights proportional to their pre-period rides per hour.

9 **Table 9** reports both, applied to the same dR series: the pre-period mean test of Zhang et al. (2026) and the
10 RDST. They ask different things of that series, whether the pre-period difference is zero and whether it drifts; the gating
11 below uses the RDST, since parallel trends restricts the drift. All five pooled RDSTs pass, as do 65 of the 75 flows;
12 the ten that fail are marked there and excluded from the interpretation in Section 7.3. The counts are a property of the
13 resolution at which the test is run rather than of the estimator: on residuals from M0 and from M1 the same RDST
14 gives 63 and 65 of 75, and the pooled test passes for all five modes under all three specifications. The pooled test is
15 itself uninformative about which flows fail.

16 7.3 Interpretation of Results

17 The estimates are read below on the flows that pass, first at the pooled level and then resolved by commute
18 structure, ring distance and the covariates.

TABLE 9 Parallel-Trends Test Results at 5%; the RDST and, for Comparison, the Pre-Period Mean Test Applied to the Same Series

mode	RDST			pre-period residual difference mean test	
	pooled	flows pass	failing flows	pooled	flows pass
Yellow Taxi	pass	14/15	0→0 (evening peak)	pass	8/15
Uber	pass	12/15	0→2 (morning peak, off-peak, evening peak)	pass	9/15
Lyft	pass	13/15	0→2 (morning peak); 2→0 (off-peak)	pass	10/15
Subway	pass	15/15	—	pass	6/15
Citi Bike	pass	11/15	0→1 (off-peak); 0→2 (off-peak); 1→0 (evening peak); 2→0 (off-peak)	pass	11/15
all modes	5/5	65/75	—	5/5	44/75

7.3.1 Pooled Policy Effect

The pooled coefficients in **Table 8** order the tolled modes. Yellow Taxi shows the deepest suppression (-20.3% , $p < 0.001$), Uber follows (-6.2% , $p = 0.049$), and Lyft is of comparable magnitude but not significant (-5.1% , $p = 0.186$). Two pre-period measures match this ordering. The first is exposure: over workday-count-weighted 2024 daily averages, 73.0% of Yellow Taxi rides fall on CRZ-touching OD ring pairs, against 33.8% for Uber and 33.9% for Lyft. The second is trip length: over CRZ-connected OD pairs the ridership-weighted mean distance is 5.06 miles for Yellow Taxi against 7.31 for Uber and 7.13 for Lyft. Because the untolled modes compete for short trips, a mode whose demand sits in the short-haul segment is the easier one to leave.

Lyft's non-significant estimate reflects dispersion rather than absence of response: its flow-level effects are more dispersed than Uber's (standard deviation 0.130 in logs against 0.101), so opposing effects offset on aggregation. The point estimate may also be attenuated: Zhang et al. (2026) report a temporary \$1.50 Lyft rider credit in January 2025 with no Uber equivalent, which would offset part of the per-trip charge over the first weeks of the post period.

Subway (-1.2% , $p = 0.700$) and Citi Bike (-3.7% , $p = 0.236$) show no pooled effect. The flow-level estimates below read this as offsetting movements across the flow dimension rather than absence of response.

7.3.2 Suppression and Substitution Under the Commute Structure

Figure 4 resolves the pooled effects onto the 15 treated flows of each mode, grouped by commute direction. The peak-direction group holds the flows running with the commute, inbound in the morning peak and outbound in the evening peak, with the two within-CRZ flows of those windows; the reverse peak-direction group holds the same ring pairs in the opposite window; the off-peak group holds every ring pair off-peak. Substitution, meaning significant positive effects on the untolled modes, is concentrated in the peak-direction group at Ring 1 or closer: three such flows for Subway and three for Citi Bike, ranging from $+5.5\%$ to $+12.8\%$. Two Subway off-peak flows on the outer ring are exceptions ($+15.1\%$ and $+17.5\%$). Suppression of the tolled modes, by contrast, is not confined to particular rings or windows: negative effects appear throughout, the deepest on Yellow Taxi $0 \rightarrow 2$ in the morning peak (-36.0%).

Within the peak-direction group the two untolled modes divide the commute by time of day. On the morning inbound flow ($1 \rightarrow 0$ morning peak) the significant gain is Subway's ($+5.5\%$), against Yellow Taxi -9.9% and Uber -7.9% , Citi Bike flat ($+0.3\%$, not significant); on the evening outbound flow ($0 \rightarrow 1$ evening peak) the gain is Citi Bike's ($+12.8\%$), against Yellow Taxi -11.8% and Uber -6.5% , with Subway not significant ($+1.9\%$). The division follows the modes' relative strengths: morning inbound trips cross a ring boundary and favour rail, evening outbound departures are freer in timing and favour cycling. Within the CRZ both untolled modes gain in both peaks, but only the morning gain has a matching suppression to set against it. In the morning peak Subway is $+8.4\%$ and Citi Bike $+6.6\%$, against Yellow Taxi -20.6% and Uber -6.2% in the same flow. In the evening peak Subway is $+9.5\%$ and Citi Bike $+11.6\%$, but no matching significant suppression is available within the modes studied here: Yellow Taxi is excluded by the RDST and Uber's estimate is not significant.

7.3.3 Spatial Decay of the Effects

Figure 5 arranges the same coefficients by ring distance. Within each mode and commute group the flow-level estimates are regressed by weighted least squares (weights $1/SD^2$) on the ring index of the flow, the non-CRZ end of

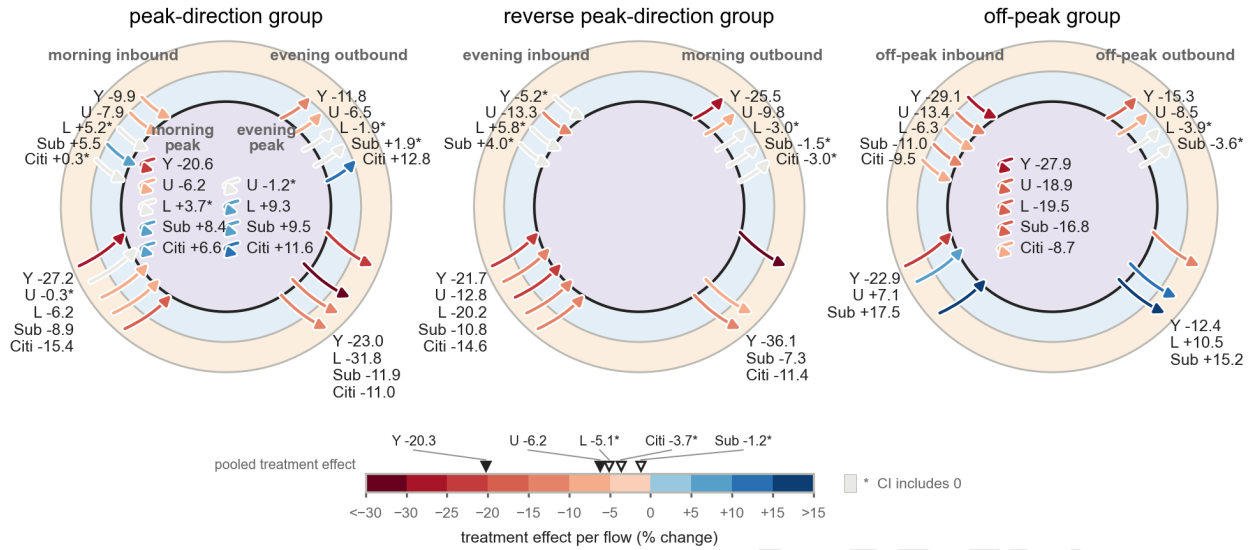


Figure 4 Treatment effect per flow, as a percentage change, on the ring cordon. Modes are abbreviated Y, U, L, Sub and Citi. Bands: Ring 0 = CRZ (purple), Ring 1 (blue), Ring 2 (orange); upper arrows Ring 1, lower arrows Ring 2, core loops within the CRZ. Triangles on the colour-bar axis give the pooled treatment effect per mode. A star marks an estimate whose 95% confidence interval includes zero.

the pair and 0 within the CRZ, and the slope tested at 5%.

The commute structure again separates the two regimes. For the tolled modes suppression weakens with ring distance in the off-peak group. The ring-mean effect narrows from -27.9% at Ring 0 to -22.2% at Ring 1 and -17.7% at Ring 2 for Yellow Taxi, from -18.9% to -11.0% to $+7.1\%$ for Uber, and from -19.5% to -5.1% to $+10.5\%$ for Lyft.

For the untolled modes the decay runs the other way, and appears in both commute groups. Along the peak direction the ring mean falls from $+9.0\%$ at Ring 0 to $+3.7\%$ at Ring 1 and -10.4% at Ring 2 for Subway, and from $+9.1\%$ to $+6.5\%$ to -13.2% for Citi Bike, so substitution not only disappears beyond Ring 1 but reverses sign. Suppression therefore attenuates spatially in the off-peak window and substitution in the peak hours.

7.3.4 Covariates

In **Figure 6** temperature is significantly positive only for Citi Bike ($+39.7\%$ per SD), the one mode whose trip is made outdoors; for the other four the journey is sheltered and the weather coefficients are correspondingly small. The weather terms enter as controls and are not read further; the selection experiments showed they leave the treatment estimate unmoved.

The AMMI coefficient is a different case. On the log scale it is far larger for Subway (2.26) and Citi Bike (1.99) than for the tolled modes (0.74–0.93), and because the score is built from pre-period ridership on the OD ring pair $\times$ peak window dimension, the large coefficients indicate that demand for the two untolled modes was already organized around the commute before the toll. That is the same dimension on which the substitution reported above is concentrated, so the two results corroborate one another.

8 CONCLUSION

A cross-classified random-effects difference-in-differences model was fit once per mode to OD ring-pair panels covering the five modes over 2024–2025, with a flow, one directed OD ring pair in one peak window, as the unit of observation. A single fit delivers the pooled effect together with the 15 flow-level effects, and equality tests confirm that it reproduces the two-way fixed-effects and fully interacted models it nests.

At the pooled level the estimates for all three tolled modes are negative, significantly so for Yellow Taxi (-20.3%) and Uber (-6.2%) but not for Lyft, while neither untolled mode changes significantly in aggregate. Where the effect is identified its size is not well explained by the fee schedule alone: Yellow Taxi pays half the per-trip charge of the ride-hailing platforms yet responds more than three times as strongly as Uber. Its exposure to the zone is roughly twice theirs and its trips are markedly shorter, so the incidence across operators is more consistent with exposure and

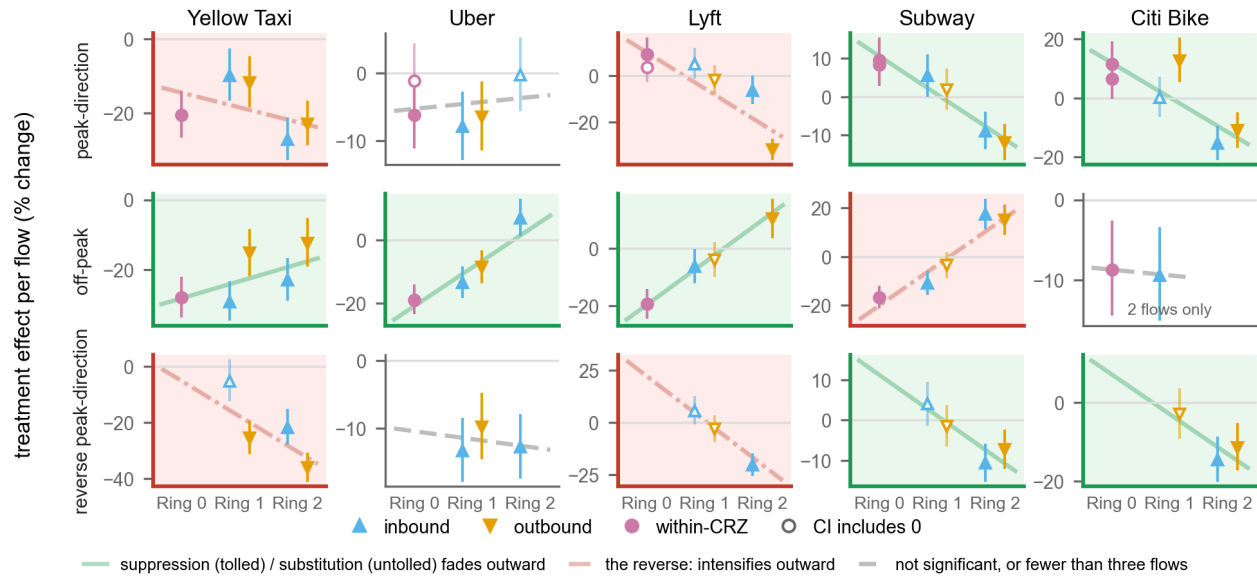


Figure 5 Treatment effect per flow, as a percentage change, with 95% confidence intervals along Ring 0 → Ring 1 → Ring 2. Lines are weighted least-squares fits on the ring index, tested at 5%; a hollow marker denotes an interval that includes zero.

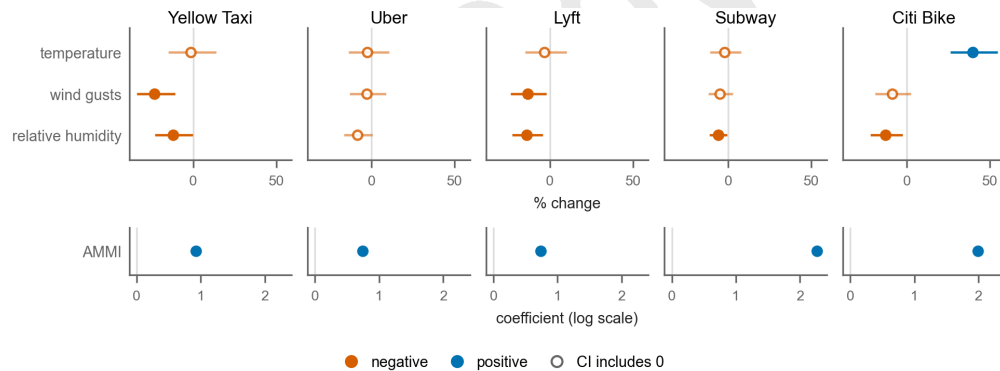


Figure 6 Covariate coefficients of M2 with 95% confidence intervals: the weather terms as a percentage change per +1 standard deviation of the z-scored covariate, the AMMI term as the coefficient itself, on the log scale, per +1 unit of the peak-direction score.

- 1 trip-length composition than with the published rate.
- 2 Once resolved to flows, the two effects have different spatiotemporal boundaries. Significant substitution falls
- 3 almost entirely in the peak-direction group at Ring 1 or closer, and the two untolled modes divide it by time of day,
- 4 Subway on the morning inbound flow and Citi Bike on the evening outbound one. Within the CRZ both gain in both
- 5 peaks, though only the morning has a matching significant suppression. Suppression shows no comparable boundary,
- 6 with negative effects across every ring and every window and the deepest on Yellow Taxi 0 → 2 in the morning peak.
- 7 Suppression and substitution also decay in space in opposite directions and in different windows. Tolled
- 8 suppression decays in the off-peak window, turning positive by Ring 2 for Uber and Lyft. Untolled substitution decays
- 9 in the peak hours and reverses sign beyond Ring 1.
- 10 Two things follow. Pricing a cordon charge by vehicle class alone leaves out much of what governs the
- 11 burden, and such a charge should also consider a mode's trip length and the share of its trips exposed to the zone, both
- 12 computable from existing OD records before the charge begins. And the untolled gains sit largely within the travel
- 13 regularity those modes already have: the AMMI coefficients show both were organized around the commute before
- 14 the toll, and their significant gains fall along it within the near ring. Beyond it their estimates turn negative alongside
- 15 the tolled ones, so the two move together rather than in opposite directions. The capacity that moves with the commute

is therefore not total capacity but the part of it already aligned with it.

One feature of the design bounds the comparison. The rings are built per mode, each from that mode's own travel linkage to the zone, and the five partitions differ over a transition band between the near and the far core, so effects are compared across modes at the same time but over nearly, not exactly, the same space. A geographic unit finer than the NTA, and a network-aware clustering that uses the road and rail structure connecting a neighborhood to the zone rather than Euclidean distance between feature vectors, would narrow that gap.

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AUTHOR CONTRIBUTIONS

The authors confirm contribution to the paper as follows: study conception and design: L. Han, X. Di; data collection: L. Han; analysis and interpretation of results: L. Han, X. Di; draft manuscript preparation: L. Han. All authors reviewed the results and approved the final version of the manuscript.

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REFERENCES

- Abadie, Alberto, Alexis Diamond, and Jens Hainmueller. 2010. "Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program." *Journal of the American Statistical Association* 105 (490): 493–505. 10.1198/jasa.2009.ap08746
- Anderson, Michael L. 2014. "Subways, Strikes, and Slowdowns: The Impacts of Public Transit on Traffic Congestion." *American Economic Review* 104 (9): 2763–2796. 10.1257/aer.104.9.2763
- Baghestani, Amirhossein, Mohammad Tayarani, Mahdiah Allahviranloo, and H. Oliver Gao. 2020. "Evaluating the Traffic and Emissions Impacts of Congestion Pricing in New York City." *Sustainability* 12 (9): 3655. 10.3390/su12093655
- Baker, Andrew, Brantly Callaway, Scott Cunningham, Andrew Goodman-Bacon, and Pedro H. C. Sant'Anna. 2026. "Difference-in-Differences Designs: A Practitioner's Guide." *Journal of Economic Literature* 64 (2): 498–557. 10.1257/jel.20251650
- Bates, Douglas, Martin Mächler, Ben Bolker, and Steve Walker. 2015. "Fitting Linear Mixed-Effects Models Using lme4." *Journal of Statistical Software* 67 (1): 1–48. 10.18637/jss.v067.i01
- Bosco, Bruno Paolo, Carlo Federico Bosco, and Paolo Maranzano. 2025. *Income Tax Treatment and Labour Supply in a Multi-Level Hierarchical Difference-in-Differences Model*. FEEM Working Paper 20.2025. Fondazione Eni Enrico Mattei. 10.2139/ssrn.5702322
- Callaway, Brantly, and Pedro H. C. Sant'Anna. 2021. "Difference-in-Differences with Multiple Time Periods." *Journal of Econometrics* 225 (2): 200–230. 10.1016/j.jeconom.2020.12.001
- Campbell, Kayleigh B., and Candace Brakewood. 2017. "Sharing Riders: How Bikesharing Impacts Bus Ridership in New York City." *Transportation Research Part A: Policy and Practice* 100: 264–282. 10.1016/j.tra.2017.04.017
- Cats, Oded, Rafał Kucharski, Santosh Rao Danda, and Menno Yap. 2022. "Beyond the Dichotomy: How Ride-Hailing Competes with and Complements Public Transport." *PLOS ONE* 17 (1): e0262496. 10.1371/journal.pone.0262496
- Citi Bike. n.d. "Citi Bike System Data." Accessed 2026. <https://citibikenyc.com/system-data>
- Cook, Cody, Aboudy Kreidieh, Shoshana Vasserman, Hunt Allcott, Neha Arora, Freek van Sambeek, Andrew Tomkins, and Eray Turkel. 2025. *The Short-Run Effects of Congestion Pricing in New York City*. NBER Working Paper 33584. National Bureau of Economic Research. 10.3386/w33584
- Craik, Lauren, and Hamsa Balakrishnan. 2023. "Equity Impacts of the London Congestion Charging Scheme: An

- 1 Empirical Evaluation Using Synthetic Control Methods.” *Transportation Research Record* 2677 (5): 1017–1029.
2 10.1177/03611981221138801
- 3 Eliasson, Jonas, Lars Hultkrantz, Lena Nerhagen, and Lena Smidfelt Rosqvist. 2009. “The Stockholm Congestion
4 Charging Trial 2006: Overview of Effects.” *Transportation Research Part A: Policy and Practice* 43 (3): 240–250.
5 10.1016/j.tra.2008.09.007
- 6 Erhardt, Gregory D., Sneha Roy, Drew Cooper, Bhargava Sana, Mei Chen, and Joe Castiglione. 2019. “Do Trans-
7 portation Network Companies Decrease or Increase Congestion?.” *Science Advances* 5 (5): eaau2670. 10.1126/sci-
8 adv.aau2670
- 9 Fageda, Xavier. 2021. “Measuring the Impact of Ride-Hailing Firms on Urban Congestion: The Case of Uber in
10 Europe.” *Papers in Regional Science* 100 (5): 1230–1254. 10.1111/pirs.12607
- 11 Fraser, Timothy, Yeonkyeong Gina Park, Danni Lu, Mohammad Tayarani, Haiyan Deng, and H. Oliver Gao. 2025.
12 “A First Look into Congestion Pricing in the United States: PM2.5 Impacts After Six Months of New York City
13 Cordon Pricing.” *npj Clean Air* 1 (1): 39. 10.1038/s44407-025-00037-2
- 14 Gauch, Hugh G. 1988. “Model Selection and Validation for Yield Trials with Interaction.” *Biometrics* 44 (3): 705–715.
15 10.2307/2531585
- 16 Ghassabian, Akhgar, Andrea R. Titus, Sarah Conderino, Alexander Azan, Rachel Weinberger, and Lorna E. Thorpe.
17 2024. “Beyond Traffic Jam Alleviation: Evaluating the Health and Health Equity Impacts of New York City’s
18 Congestion Pricing Plan.” *Journal of Epidemiology and Community Health* 78 (5): 273–276. 10.1136/jech-2023-
19 221639
- 20 Gibson, Matthew, and Maria Carnovale. 2015. “The Effects of Road Pricing on Driver Behavior and Air Pollution.”
21 *Journal of Urban Economics* 89: 62–73. 10.1016/j.jue.2015.06.005
- 22 Green, Colin P., John S. Heywood, and Maria Navarro Paniagua. 2020. “Did the London Congestion Charge Reduce
23 Pollution?.” *Regional Science and Urban Economics* 84: 103573. 10.1016/j.regsciurbeco.2020.103573
- 24 Hall, Jonathan D., Craig Palsson, and Joseph Price. 2018. “Is Uber a Substitute or Complement for Public Transit?.”
25 *Journal of Urban Economics* 108: 36–50. 10.1016/j.jue.2018.09.003
- 26 Herzog, Ian. 2024. “The City-Wide Effects of Tolling Downtown Drivers: Evidence from London’s Congestion
27 Charge.” *Journal of Urban Economics* 144: 103714. 10.1016/j.jue.2024.103714
- 28 Jin, Scarlett T., Hui Kong, and Daniel Z. Sui. 2019. “Uber, Public Transit, and Urban Transportation Equity: A Case
29 Study in New York City.” *The Professional Geographer* 71 (2): 315–330. 10.1080/00330124.2018.1531038
- 30 Krishnamurthy, Chandra Kiran B., and Nicole S. Ngo. 2024. “Do Ride-Hailing Services Worsen Freeway Congestion
31 and Air Quality? Evidence from Uber’s Entry in California.” *Journal of the Association of Environmental and*
32 *Resource Economists* 11 (6): 1559–1603. 10.1086/729901
- 33 Kuznetsova, Alexandra, Per B. Brockhoff, and Rune H. B. Christensen. 2017. “lmerTest Package: Tests in Linear
34 Mixed Effects Models.” *Journal of Statistical Software* 82 (13): 1–26. 10.18637/jss.v082.i13
- 35 Leape, Jonathan. 2006. “The London Congestion Charge.” *Journal of Economic Perspectives* 20 (4): 157–176.
36 10.1257/jep.20.4.157
- 37 Li, Donghang, Dingyi Zhuang, Yunlin Li, Chenan Shen, Nina Cao, Yunhan Zheng, Shenhao Wang, and Jinhua
38 Zhao. 2026. “Public Transit Gains and Spatially Uneven Travel Demand Changes After NYC Congestion Pricing.”
39 arXiv:2606.17530. 10.48550/arXiv.2606.17530
- 40 Li, Sen, Kameshwar Poolla, and Pravin Varaiya. 2021. “Impact of Congestion Charge and Minimum Wage on
41 TNCs: A Case Study for San Francisco.” *Transportation Research Part A: Policy and Practice* 148: 237–261.
42 10.1016/j.tra.2021.02.008
- 43 Li, Yanchao, and Daniel Vignon. 2024. “Do Ride-Hailing Congestion Fees in NYC Work?.” *Transportation Research*
44 *Part A: Policy and Practice* 190: 104274. 10.1016/j.tra.2024.104274
- 45 Liang, Yuan, Bingjie Yu, Xiaojian Zhang, Yi Lu, and Linchuan Yang. 2023. “The Short-Term Impact of Congestion
46 Taxes on Ridesourcing Demand and Traffic Congestion: Evidence from Chicago.” *Transportation Research Part A:*
47 *Policy and Practice* 172: 103661. 10.1016/j.tra.2023.103661
- 48 Metropolitan Transportation Authority. 2023. “MTA Subway Origin–Destination Ridership Estimate: 2023.” New
49 York State Open Data. <https://data.ny.gov/d/uhf3-t34z>
- 50 Metropolitan Transportation Authority. 2024. “MTA Subway Origin–Destination Ridership Estimate: 2024.” New
51 York State Open Data. <https://data.ny.gov/d/jsu2-fbtj>
- 52 Metropolitan Transportation Authority. 2025. “MTA Subway Origin–Destination Ridership Estimate: 2025.” New
53 York State Open Data. <https://data.ny.gov/d/y2qv-fytt>
- 54 Metropolitan Transportation Authority. n.d.a. “Central Business District Tolling Program: The Congestion Relief

- Zone.” Accessed 2026. <https://www.mta.info/project/CBDTP>
- Metropolitan Transportation Authority. n.d.b. “MTA General Transit Feed Specification (GTFS) Static Data.” New York State Open Data. Accessed 2026. <https://data.ny.gov/d/fgm6-ccue>
- New York City Department of City Planning. 2020. “2020 Neighborhood Tabulation Areas (NTAs).” NYC Open Data. <https://data.cityofnewyork.us/d/9nt8-h7nd>
- New York City Department of City Planning. n.d. “Primary Land Use Tax Lot Output (PLUTO).” NYC Open Data. Accessed 2026. <https://data.cityofnewyork.us/d/64uk-42ks>
- New York City Taxi and Limousine Commission. n.d.a. “TLC Trip Record Data.” Accessed 2026. <https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>
- New York City Taxi and Limousine Commission. n.d.b. “NYC Taxi Zones.” NYC Open Data. Accessed 2026. <https://data.cityofnewyork.us/d/8meu-9t5y>
- Ostrovsky, Michael, and Frank Yang. 2025. *Effective and Equitable Congestion Pricing: New York City and Beyond*. Working Paper 4228. Stanford University Graduate School of Business. Version of July 1, 2025. <https://web.stanford.edu/~ost/papers/nyc20250701.pdf>
- R Core Team. 2025. “R: A Language and Environment for Statistical Computing.” <https://www.R-project.org/>
- Raudenbush, Stephen W. 1993. “A Crossed Random Effects Model for Unbalanced Data with Applications in Cross-Sectional and Longitudinal Research.” *Journal of Educational Statistics* 18 (4): 321–349. 10.3102/10769986018004321
- Raudenbush, Stephen W., and Anthony S. Bryk. 2002. *Hierarchical Linear Models: Applications and Data Analysis Methods*, 2nd ed. Thousand Oaks, CA: Sage Publications. Chapter 12, Models for Cross-Classified Random Effects.
- Ren, Xiyuan, Zhenglei Ji, and Joseph Y. J. Chow. 2026. “Distributional Welfare Impacts and Compensatory Transit Strategies under NYC Congestion Pricing.” *Transportation Research Part A: Policy and Practice* 212: 105159. 10.1016/j.tra.2026.105159
- Rosaia, Nicola. 2025. “Competing Platforms and Transport Equilibrium.” *Econometrica* 93 (6): 2235–2271. 10.3982/ECTA21773
- Shanbhag, Nikhil, Joshua Wu, and Xuan Di. 2026. “Causal Effects of Congestion Pricing on Urban Mobility by Neighborhood: A Generalized Synthetic Control Approach.” Paper presented at the 2026 IEEE International Conference on Intelligent Transportation Systems (ITSC), Naples, Italy. Accepted. 10.2139/ssrn.6512662
- Simeonova, Emilia, Janet Currie, Peter Nilsson, and Reed Walker. 2021. “Congestion Pricing, Air Pollution, and Children’s Health.” *Journal of Human Resources* 56 (4): 971–996. 10.3368/jhr.56.4.0218-9363r2
- Tarduno, Matthew. 2021. “The Congestion Costs of Uber and Lyft.” *Journal of Urban Economics* 122: 103318. 10.1016/j.jue.2020.103318
- Turkcan, Mehmet Kerem, Jhonatan Tavori, Javad Ghaderi, Gil Zussman, Zoran Kostic, and Andrew Smyth. 2026. “A Vision-Based Analysis of Congestion Pricing in New York City.” arXiv:2602.03015. 10.48550/arXiv.2602.03015
- Vacanza Team. 2025. “holidays: Open World Holidays Framework.” Python package, version 0.100. <https://pypi.org/project/holidays/>
- Vickrey, William S. 1969. “Congestion Theory and Transport Investment.” *American Economic Review* 59 (2): 251–260. <https://www.jstor.org/stable/1823678>
- Virtanen, Pauli, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, et al. 2020. “SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python.” *Nature Methods* 17: 261–272. 10.1038/s41592-019-0686-2
- Wang, Mingyin, and Xuan Di. 2026. “Has Congestion Pricing Improved Short-Term Road Safety? A Case Study in New York City.” *Safety* 12 (3): 64. 10.3390/safety12030064
- Watson, Peter L., and Edward P. Holland. 1976. “Congestion Pricing: The Example of Singapore.” *Finance and Development* 13 (1): 20–23. <https://www.elibrary.imf.org/view/journals/022/0013/001/article-A008-en.xml>
- Xu, Yiqing. 2017. “Generalized Synthetic Control Method: Causal Inference with Interactive Fixed Effects Models.” *Political Analysis* 25 (1): 57–76. 10.1017/pan.2016.2
- Zhang, Yueshan, Yanling Sang, and Manxi Wu. 2026. “Congestion Pricing in New York City: Effects on Ride-Hailing and Transit.” *Transportation Research Part A: Policy and Practice* 208: 104966. 10.1016/j.tra.2026.104966
- Zheng, Yunhan, Patrick Meredith-Karam, Anson Stewart, Hui Kong, and Jinhua Zhao. 2023. “Impacts of Congestion Pricing on Ride-Hailing Ridership: Evidence from Chicago.” *Transportation Research Part A: Policy and Practice* 170: 103639. 10.1016/j.tra.2023.103639
- Zippenfenig, Patrick. 2023. “Open-Meteo.com Weather API.” Zenodo. 10.5281/zenodo.7970649